# Stav Grossfeld


- stav.grossfeld@gmail.com

**Linkedin:** https://www.linkedin.com/in/stav-grossfeld/ **Github:** https://github.com/stavgrossfeld


**Programming Languages:** Python (Scikit-learn, Pandas, NumPy, Pytorch), R (Tidyverse, Dplyr), Nextflow, Bash, SQL

**Data Visualization:** R (ggplot2, Shiny), Python (Matplotlib, Seaborn, Plotly), Tableau

**Bioinformatics Tools:** Samtools, GATK, Bedtools, VEP, snpEff

**Cloud & HPC Technologies:** AWS, Slurm HPC, Docker, Linux, Git

**Data Engineering & Workflow Tools:** Nextflow, Hive, Hadoop, Dagster, Athena, Glue, LakeFS

**Analyses & Methods:** GWAS, differential expression (DESeq), gene set enrichment analysis (GSEA), Single-cell analysis, Dimensionality Reduction (PCA, UMAP, t-SNE), Machine Learning (supervised/unsupervised, Deep Learning), Spatial Transcriptomics

**Genetic Databases:** cBioPortal, ClinVar, CPTAC, dbSNP, DepMap, GEO, gnomAD, HapMap, OMIM, TCGA, UCSC

## Working experience:

**Associate Bioinformatics Engineer – Arcus Biosciences (September 2022 - Present):**

I. Pipeline Engineering & Analysis

- Architected and engineered a fit-for-purpose downstream nextflow pipeline for nf-core/sarek to generate consensus variant calls and extract key tumor instability metrics, directly informing clinical trial decision-making.
- Developed a rigorous data-validation framework to visualize mutation overlap across pipeline versions, serving as an E2E diagnostic tool to ensure 100% data consistency and integrity for clinical trials.
- Spearheaded the development of internal Nextflow workflows and benchmarking strategies, serving as the lead technical resource for nf-core best practices and SOPs.
- Optimized high-performance workflows on AWS and a 2400-core Slurm HPC.
- Performed bespoke variant-calling analysis and deep-dive troubleshooting for non-standard clinical samples, utilizing custom Python and R scripts to resolve data discrepancies and salvage high-value trial insights.
- Conducted ad-hoc, one-off RNA-seq analysis projects to investigate specific biological hypotheses, delivering rapid turnaround on exploratory research questions

II. Infrastructure & Automation

- Implemented automated CI/CD pipelines and unit testing suites to ensure the reliability and reproducibility of production-level bioinformatics code.
- Built a multimodal data lake architecture using Dagster and LakeFS, enabling version-controlled ingestion of S3 biological data and streamlined querying via AWS Athena.
- Developed and implemented benchmarking strategies to quantitatively evaluate and improve the performance, cost-efficiency, and reliability of internal computational tools.

III. Data Products & Cross-Functional Impact

- Designed an R package and custom R Shiny dashboards (e.g., CPTAC correlation tool) to provide on-demand genetic analysis and visualization for cross-functional R&D teams.
- Developed a complex sample manifest platform using Shiny, architecting a multi-page sequencing submission system that streamlined tracking and validation across multiple external vendors.
- Collaborated with cross-functional teams to design and build scalable infrastructure for managing complex multimodal bioinformatics and clinical datasets.

*** References available upon request ***

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**Bioinformatics Scientist - Phosphorus (January 2021 – June 2022):** Preventative medical genetics testing

- Designing, developing, and curating NGS analysis pipelines running on AWS for patient samples (alignment, variant calling, etc) processing genetic data and storing information in databases using Python, Bash, SQL, GATK, and Ruby.
- Proposing new directions and methods for new bioinformatics pipelines based on needs for new information in patient reports and internal software platform.
- Monitoring new bioinformatics tools and research to adopt and implement advanced solutions.
- Troubleshooting pipelines, curating and diagnosing data issues and implementing fixes as needed.
- Importing, transforming, and connecting information from public databases such as Gnomad, OMIM, and Clinvar. **Publications:** **Technical Performance of a 430-Gene Preventative Genomics Assay to Identify Multiple Variant Types Associated with Adult-Onset Monogenic Conditions, Susceptibility Loci, and Pharmacogenetic Insights** (Apr. 2022)

**Bioinformatics Intern - Biomarin (June 2020 - November 2020):** Human genetics group

- Implementing machine learning algorithms to characterize alternative splicing of public RNA seq datasets using GATK, R, Keras, Python, Bash.
- Proposing new directions and methods for investigation based on independent literature reviews and analysis of relevant research topics, and effectively presenting findings to multiple types of audiences.

**Bioinformatician - Translational Genomics USC (August 2019 - June 2020):** PI. Dr. Brooke Hjelm's lab

- Worked on multiple projects involving pathway analysis in complex neurological disorders using R, Python, Bash, and bioinformatics toolkits such as Samtools and GATK.
- Modeling neuronal activity-induced myelination in RNA-Seq datasets across multiple disorders.
- Spatial transcriptomics analysis using data from the 10x Genomics Visium Gene Expression Solution.
- Evaluation of Copy Number Variants in Psychiatric Disorder Brains and Methods Development for Sample Identification in sc-CNV Data.

**Data Scientist - Tailormed (July 2018 - May 2019):** Medical Financial Startup optimizing financial navigation for cancer patients in the American health system.

- Defined the scope of problems, chose approaches and processes for applying Machine Learning on complex medical data. Prototyped and implemented machine learning algorithms, communicated findings and results to stakeholders.
- Engineered pipelines using SQL, Python, Node.js connecting and transforming multiple data sources, managed production runs on AWS and data transforms for storage and machine learning feeds.

**Data Analyst - Scribd San Francisco (January 2017- 2017):** Subscription Ebook / Audiobook Startup with over 500k monthly paying subscribers

- Engineered large-scale ETL pipelines and complex dependency graphs using SQL and Python to automate daily data extraction and transformation from SQL to partitioned HDFS tables.
- Conducted A/B test analyses and architected automated reporting solutions, including an "Invalid Events" dashboard, to drive UX optimizations and streamline cross-functional business intelligence.

## Education:

**USC Keck School of Medicine, M.S. Translational Biomedical Informatics (2020)**

**UC Davis, B.S. in Genetics;Technology Management Minor (2015)**

*** References available upon request ***